

Relational Database Systems (RDS)

Class no: 24

What is AWS RDS?

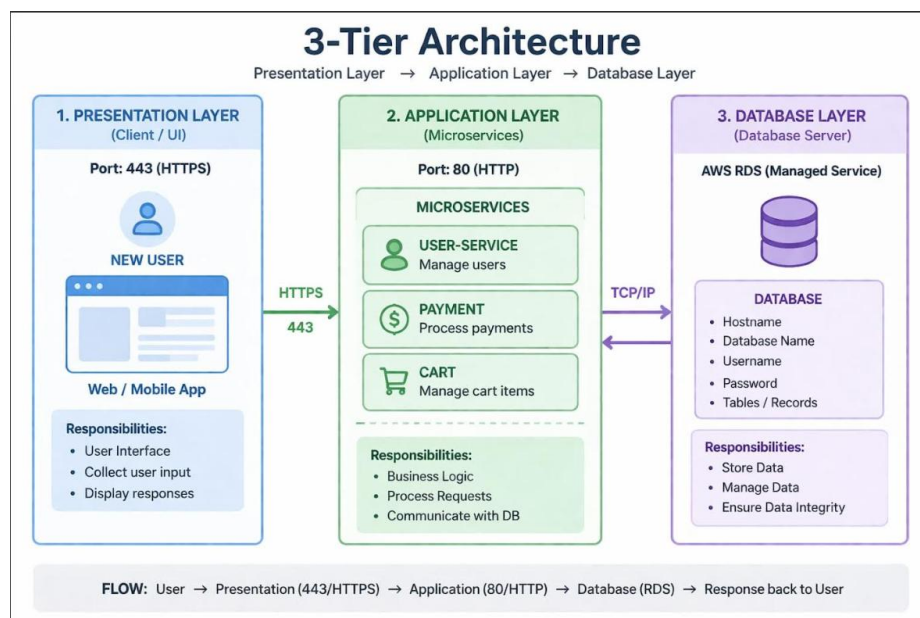
Amazon Relational Database Service (RDS) is a fully managed cloud service provided by AWS to simplify the setup, operation, and scaling of relational databases. It abstracts and automates many database administration tasks, such as:

- Hardware provisioning
- Database installation and configuration
- Patch management
- Automated backups and snapshots
- Monitoring and alerting
- Failover and disaster recovery

This allows developers and businesses to focus on application development rather than database maintenance, while AWS manages infrastructure and routine administrative tasks.

Key Advantages:

- Cost-efficient, resizable compute and storage capacities
- High availability via Multi-AZ deployments
- Automatic backups and point-in-time recovery
- Scalability through vertical scaling and read replicas
- Security using IAM, VPC isolation, SSL, and KMS encryption



Databases Supported by RDS

AWS RDS supports six major relational database engines, each offering distinct features:

1. Amazon Aurora

- MySQL and PostgreSQL compatible, cloud-optimized, high performance
- Automatically replicates data across multiple Availability Zones

2. PostgreSQL

- Advanced open-source database
- Supports complex queries, large datasets, GIS, and analytics applications

3. MySQL

- Widely used open-source relational database
- Well-suited for web applications, supports automated backups and Multi-AZ

4. MariaDB

- Open-source fork of MySQL, compatible with existing MySQL applications
- Offers additional storage engines and features for enhanced performance

5. Oracle Database

- Enterprise-grade commercial database
- Advanced SQL, PL/SQL, analytics, and robust transaction management
- Licensing options: License Included (LI) or Bring Your Own License (BYOL)

6. Microsoft SQL Server

- Popular enterprise database for Windows environments
- Supports multiple editions: Express, Web, Standard, Enterprise
- Features include TDE, replication, advanced reporting, and analytics

Note: Each engine comes with different management, parameter, and option groups that specify the behavior of the database. Features may vary by engine, version, and region.

Default Port Numbers for RDS Databases

AWS RDS uses standard TCP ports for database connections, which can usually be customized if needed:

Database Engine	Default Port
PostgreSQL	5432

Database Engine	Default Port
MySQL / MariaDB	3306
Oracle Database	1521
Microsoft SQL Server	1433
Amazon Aurora	3306 (MySQL-compatible) or 5432 (PostgreSQL-compatible)

These ports must be opened in the RDS instance security group to allow client connections.

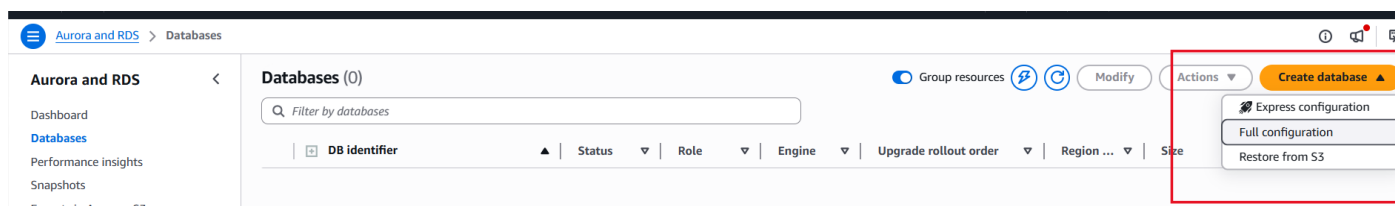
Hands on: Creating a RDS and connecting with an EC2 instance.

Step no:1 Search RDS → click Aurora and RDS.



Choose create Database → Full Configuration.

Express configuration is the database created automatically. Full configuration is where we choose and personalize the database.



Step no:2 Choose MySQL,

Method: **Full configuration**

Template: Currently we are using free tier.

It normally depends on the environment we are creating the database like development, production.

Availability and durability: It depends on our need. More AZs more availability.

Create database [Info](#)

Free plan has access to limited features and resources

The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)



☐ Aurora (PostgreSQL Compatible)



☒ MySQL



☐ MariaDB



☐ Oracle

ORACLE

☐ Microsoft SQL Server



Choose a database creation method

☒ Full configuration

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create

Use recommended best-practice configurations. Some configurati

Templates

Choose a sample template to meet your use case.

☐ Production

Use defaults for high availability and fast, consistent performance.

☐ Dev/Test

This instance is intended for development use outside of a production environment.

☒ Free tier

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Availability and durability

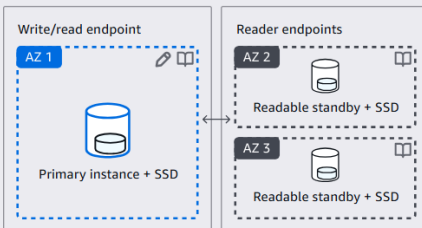
Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

☐ Multi-AZ DB cluster deployment (3 instances)

Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

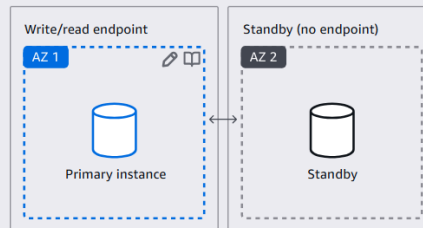
- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency



☐ Multi-AZ DB instance deployment (2 instances)

Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

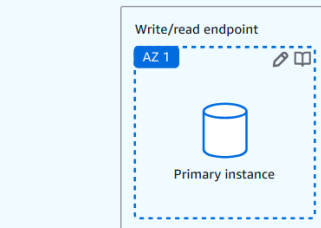
- 99.95% uptime
- Redundancy across Availability Zones



☒ Single-AZ DB instance deployment (1 instance)

Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy



Step no:3 Enable → MySQL community.

Engine version: **Default (8.4.8)**

Disable: **RDS extension support**

DB- instance identifier: We provide the name of the name to identify. → Here it is **database-1**.

Settings

Edition

☒ MySQL Community

Engine version [Info](#)

View the engine versions that support the following database features.

▼ Hide filters

- ☐ Show only versions that support the Multi-AZ DB cluster [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster trans.
- ☐ Show only versions that support the Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

MySQL 8.4.8

- ☐ Enable RDS Extended Support [Info](#)
Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your data major version in the [RDS for MySQL documentation](#).

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

database-1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. FI

Credential settings,

Master username: **admin** (can perform READ and WRITE operations in database)

Password: **Self managed** → Create a password which does not contain any special characters.

Database authentication as password authentication.

(this username and password is used to connect with the microservices of the application)

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

- ☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
- ☒ Self managed
Create your own password or have RDS create a password that you manage.
- ☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength

Average

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

▼ Additional credentials settings

Database authentication options [Info](#)

- ☒ Password authentication
Authenticates using database passwords.
- ☐ Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.
- ☐ Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Instance configuration,

Instance type: **t3.micro** (As it is a database EC2 server)

Storage type: **General purpose SSD**

Allocated storage: **30**

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class | [Info](#)

▼ Hide filters

- ☐ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.
- ☐ Include previous generation classes
- ☐ Standard classes (includes m classes)
- ☐ Memory optimized classes (includes r and x classes)
- ☒ Burstable classes (includes t classes)

Instance type

db.t3.micro

2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)

Baseline performance determined by volume size

Allocated storage [Info](#)

30

Allocated storage value must be 20 GiB to 6,144 GiB

▼ Additional storage configuration

Storage autoscaling [Info](#)

Provides dynamic scaling support for your database's storage based on your application's needs.

- ☐ Enable storage autoscaling
Enabling this feature will allow the storage to increase after the specified threshold is exceeded.

Connectivity,

Compute resource: we will create an attach an EC2 later. Hence **Don't connect**.

VPC: **Default VPC**

Subnet: **default**

Enable: **Public access**

Security group: **Default** (ensure ALL TCP and MYSQL (3306) is added in inbound rules)

Choose the AZs

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-0538c231a226fe3c9)
3 Subnets, 3 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access [Info](#)

☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resource to the database.

☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resource to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups

Choose one or more options

default X

Availability Zone [Info](#)

ap-south-1a

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ **Create an RDS Proxy** [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically

rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061

If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

Database port [Info](#)
TCP/IP port that the database will use for application connections.

3306

Monitoring,

We can choose log reports to monitor the database and use it for debugging purpose.

Monitoring [Info](#)

Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. **Database Insights pricing** [CloudWatch pricing](#) [↗](#).

☐ Database Insights - Advanced

- Retains 15 months of performance history
- Fleet-level monitoring
- Integration with CloudWatch Application Signals

☒ Database Insights - Standard

▼ Additional monitoring settings

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

☐ Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

- ☒ Audit log
- ☒ Error log
- ☒ General log
- ☒ iam-db-auth-error log
- ☒ Slow query log

IAM role

The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

Click **Create database**.

Step no:4 Once created Click it, we can see the code to connect with an EC2 instance using the curl as well as mysql.

database-1

Summary

DB identifier
database-1

CPU
 20.82%

Status

 Configuring-enhanced-monitoring

Class

db.t3.micro

Role

Instance

Current activity

 0 Connections

Engine

MySQL Community

Region & AZ

ap-south-1a

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Connect using [Info](#)

☒ Code snippets

Use when connecting through SDK, APIs, or third-party tools including agents.

☐ CloudShell

Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☐ Endpoints

Use when connecting through endpoints.

Internet access gateway

☐ Disabled

IAM Authentication

☐ Disabled

Programming language

MySQL (macOS) ▼

Endpoint type

Instance endpoint ▼

Connection steps

Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

```
1 | curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
```

```
1 | mysql -h database-1.cbwcwsiqgdd.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY --ssl-ca=./global-bundle.pem
```


Step no:5 Create an Instance for connecting RDS Database.

Name: **My-RDS-Server**

OS type: **Ubuntu**

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

My-RDS-Server

[Add additional tags](#)

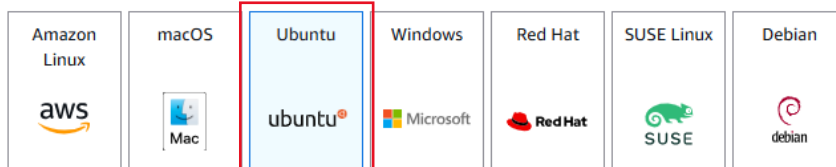
▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start



Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-05d2d839d4f73aafb (64-bit (x86)) / ami-0f6c99577b93a642b (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

AMI ID

Publish Date

Username ⓘ

64-bit (x86) ▼

ami-05d2d839d4f73aafb

2026-03-14

ubuntu

Verified provider

Instance type: **t3.micro**

Keypair: **Provide an available one or create a new one.**

Network settings,

SG: Default SG

VPC: Default VPC

(ensure ALL TCP and MYSQL (3306) is added in inbound rules)

Launch inatance.

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0112 USD per Hour
On-Demand SUSE base pricing: 0.0112 USD per Hour On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

mumkp

[Create new key pair](#)

▼ Network settings [Info](#)

[Edit](#)

Network [Info](#)

vpc-0538c231a226fe3c9

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups

[Compare security group rules](#)

default sg-06d80e515c32b9515 ✕
VPC: vpc-0538c231a226fe3c9

Security groups that you add or remove here will be added to or removed from all your network interfaces.

To add ALL TCP (0-65535) and for connection with MYSQL server enable it (default port number:3306) in the SG inbound rules.

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID

	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-0ff6791321f332304	HTTPS	TCP	443	Custom	Q	Delete
sgr-06b9fe308e8427a99	All traffic	All	All	Custom	0.0.0.0/0 ✕	Delete
sgr-08197f521eb29746a	SSH	TCP	22	Custom	Q	Delete
sgr-0e4b541e985a71540	All ICMP - IPv4	ICMP	All	Custom	0.0.0.0/0 ✕	Delete
sgr-0f84d83f6745d9a22	HTTP	TCP	80	Custom	Q	Delete
sgr-02b0bb72fe6f5dc94	All TCP	TCP	0 - 65535	Custom	0.0.0.0/0 ✕	Delete
-	MYSQL/Aurora	TCP	3306	Anywh...	Q 0.0.0.0/0 ✕	Delete

[Add rule](#)

Step no:6 Once the EC2 instance is created → select and click connect.

In order to connect with the RDS Of MYSQL type we need to install the MYSQL server.

Switch user: **sudo su**

Update the server: **apt update -y**

```
The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>"
See "man sudo_root" for details.

ubuntu@ip-172-31-35-203:~$ sudo su
root@ip-172-31-35-203:/home/ubuntu# apt update -y
```

i-0ba4c2d1a6ea3ef0b (My-RDS-Server)

PublicIPs: 13.233.124.36 PrivateIPs: 172.31.35.203

Once updated, install MYSQL by: **apt install mysql-server**

```
root@ip-172-31-35-203:/home/ubuntu# apt install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libe
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 li
  mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libdata-dump-perl libipc-sharedcache-perl libio-compress-brotli-perl lib
The following NEW packages will be installed:
  libcgi-fast-perl libcgi-pm-perl libclone-perl libencode-locale-perl libe
  libhttp-message-perl libio-html-perl liblwp-mediatypes-perl libmecab2 li
  mysql-server mysql-server-8.0 mysql-server-core-8.0
0 upgraded, 28 newly installed, 0 to remove and 44 not upgraded.
Need to get 29.6 MB of archives.
After this operation, 243 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 mys
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main s
```

Once installed,

Copy the **mysql** command in the RDS database and paste it. (upto -p)

It will ask for a password. Enter the **password** we have created for the Master: **admin**.

Now the instance will be connected with the RDS: **database-1**.

Enter the database command: **SHOW DATABASES**. It is to see the available databases.

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-35-203:/home/ubuntu# mysql -h database-1.cbwcwsiqgdd.ap-south-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 32
Server version: 8.4.8 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.00 sec)
```

To create a database: **CREATE DATABASE database_name;**

To create a table: **CREATE TABLE table_name (Field_name, Type)**

```
mysql> CREATE DATABASE testdb;
Query OK, 1 row affected (0.03 sec)

mysql> USE testdb;
Database changed
mysql> CREATE TABLE Persons (
->   PersonID int PRIMARY KEY,
->   LastName varchar(255) NOT NULL,
->   FirstName varchar(255),
->   Address varchar(255),
->   City varchar(255)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> SHOW DATABASES
-> ^C
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| testdb |
+-----+
5 rows in set (0.00 sec)

mysql>
```

i-0ba4c2d1a6ea3ef0b (My-RDS-Server)

PublicIPs: 13.233.124.36 PrivateIPs: 172.31.35.203

Normally, we will create a database with host, user name and password. Developer will use this in the micro services to connect with our database.

MASTER and READ Replicas

In Amazon RDS (Relational Database Service), the architecture of **Primary (Master)** and **Read Replicas** is designed to separate the "writing" of data from the "reading" of data. This improves performance and ensures your application stays snappy even under heavy traffic.

1. The Primary (Master) Instance

Think of the Master as the "**Source of Truth.**" It is the main database instance that handles all the heavy lifting for data modifications.

- **Role:** Handles all **Write** operations (INSERT, UPDATE, DELETE) and **Read** operations.
- **Quantity:** There is only **one** Primary instance in a standard RDS setup.
- **Storage:** It holds the "Gold Copy" of your data.
- **Configuration:** Usually deployed in a **Multi-AZ** (Availability Zone) setup for high availability. If the Master fails, RDS automatically flips a switch to a standby instance.

2. Read Replicas

Read Replicas are "copies" of the Master instance used to offload work.

- **Role:** Handles **Read-only** queries (SELECT statements). You cannot write data directly to a replica.
- **Replication Type:** It uses **Asynchronous Replication**. This means there is a tiny delay (usually milliseconds) between data being written to the Master and appearing on the Replica.
- **Scaling:** You can create up to **15** replicas for most engines (like MySQL, PostgreSQL, and MariaDB) to handle massive amounts of read traffic.
- **Promotion:** If the Master dies and you don't have Multi-AZ, or if you just want to split the DB into two independent systems, you can **promote** a Read Replica to become its own standalone Master.

When should you use them?

- **Heavy Reporting:** If your marketing team is running huge data reports that slow down the app, point their tools to a Read Replica.
- **Global Reach:** You can place Read Replicas in different **Regions** (e.g., a Master in US-East and a Replica in Tokyo) to reduce latency for international users.
- **Business Continuity:** While not a "backup" per se, they provide an extra layer of data redundancy.

Key Differences

Feature	Primary (Master)	Read Replica
Operations	Read & Write	Read-Only
Purpose	Data Integrity & Processing	Scalability & Performance
Replication	N/A	Asynchronous
Endpoint	Unique "Write" Endpoint	Unique "Read" Endpoint(s)
Billing	Standard RDS Rate	Billed as a separate instance

Hands on: Creating a read replica for an RDS database:

Step no: 1 Choose the created database → **actions** → **create read replica**.

The screenshot shows the AWS Management Console interface for RDS databases. At the top, there's a search bar and a 'Group resources' toggle. Below, a table lists databases. The first row shows 'database-1' with status 'Available', engine 'MySQL Co...', and instance class 'db.t3.micro'. The 'Actions' button for this database is highlighted, and a dropdown menu is open, showing various actions. The 'Create read replica' option is highlighted in blue.

DB identifier	Status	Role	Engine	Upgrade rollout order	Region	Size
database-1	Available	Instance	MySQL Co...	SECOND	ap-south-1a	db.t3.micro

- Stop temporarily
- Reboot
- Delete
- Set up EC2 connection
- Set up Lambda connection
- Migrate data from EC2 database
- Create read replica**
- Create Aurora read replica
- Create blue/green deployment
- Promote
- Convert to Multi-AZ deployment
- Take snapshot
- Restore to point in time
- Migrate snapshot
- Create zero-ETL integration
- Create RDS Proxy
- Create ElastiCache cluster

Source: Our database named, **database-1**

DB instance identifier: Name of our read replica, **Mynewddb1**

Rest as default → Click **Create Read Replica**.

Create read replica

You are creating a replica DB instance from a source DB instance. This new DB instance will have the source DB instance's DB security groups and DB parameter groups.

1

Free plan has access to limited features and resources

The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

Settings

Replica source

Source DB instance identifier

database-1

Role: Instance

DB instance identifier

This is the unique key that identifies a DB instance. This parameter is stored as a lowercase string (for example, mydbinstance).

Mynewdb1

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class

Info

▼ Hide filters

☒ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

Instance type

db.t3.micro

2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Backup

Tags

☐ Copy tags to snapshots

☒ Enable encryption

Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. Info

AWS KMS key

Info

Enter a key ARN

Amazon Resource Name (ARN)

arn:aws:kms:ap-south-1:314383175192:key/65acb271-1648-4a80-ae32-10406ed9501b

Example: arn:aws:kms:<region>:<accountID>:key/<key-id>

Account

314383175192

KMS key ID

65acb271-1648-4a80-ae32-10406ed9501b

Maintenance

Auto minor version upgrade

Info

☒ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version documentation

☐ Enable deletion protection

Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Cancel

Create read replica

Once initiated we can see the replica under the respective database with status **creating**.

Databases (2) Group resources

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size
database-1	Modify...	Primary	MySQL Co...	SECOND	ap-south-1a	db.t3.micro
mynewdb1	Creating	Replica	MySQL Co...	SECOND	-	db.t3.micro

Once created, we can see the status as **available**.

Databases (2) Group resources Modify Actions Create database

Filter by databases

< 1 > ⚙

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendations	CPU	Current activity
database-1	Available	Primary	MySQL Co...	SECOND	ap-south-1a	db.t3.micro		3.83%	1 Connections
mynewdb1	Available	Replica	MySQL Co...	SECOND	ap-south-1b	db.t3.micro		4.74%	0 Connections